03 21826 Autumn 2025
LH The Life and Death of Stars

Lecture 16

(JCD 12.1 - 12.2)

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Previous lecture

- When a star starts burning H, it sits on the ZAMS and evolves in the HR diagram till the TAMS
- During core H burning, luminosity and radius will generally increase. Effective temperature can increase or decrease depending on stellar mass
- The change in hydrogen abundance depends on how radiation is transported: gradual change in radiative zones and uniform change in convective zones
- Main sequence lifetime is decreasing function of mass

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Outline

Evolution after main sequence

Moderate-mass stellar evolution
- The Schönberg-Chandrasekhar limit
- The Hertzsprung gap

Summary

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Evolution after main sequence - He core

- Main sequence stops when hydrogen is exhausted near the centre
- Core is now made of helium and some heavier elements
- Core temperature not yet high enough for helium fusion to start (inert core)
- Hence no nuclear energy generation:

$$\epsilon = 0 \Rightarrow \frac{dL}{dr} = 0 \quad \& \quad L \quad \text{is} \quad \text{small} \Rightarrow \left|\frac{dT}{dr}\right| \quad \text{is} \quad \text{small}$$

- Core almost isothermal

Evolution after main sequence - H shell fusion

- In a shell around the core, there still is hydrogen and also high enough temperature for it to fuse
- This H shell burning mainly provides the energy and thus luminosity of the star
- H shell burning leads to the mass of the He core growing

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Evolution after main sequence - core contraction

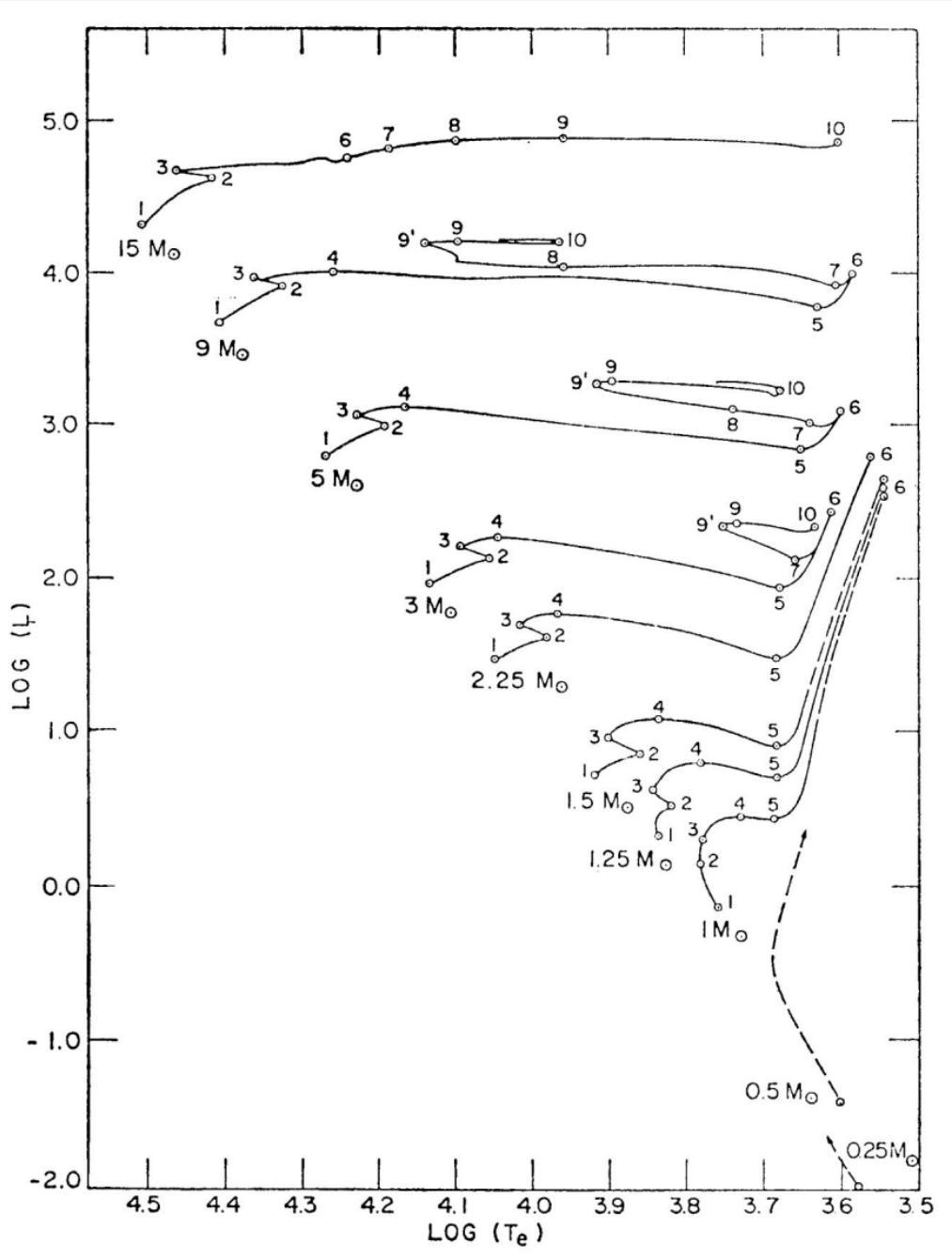
- Growing core mass leads to core contraction
- This releases gravitational potential energy
- This increases the thermal energy and thus temperature of the core (remember the virial theorem, lecture 6)
- At some point the temperature is high enough for He fusion to start

Evolution after main sequence - repeat

- The process is then roughly repeated
- The process only happens as long as the core is not degenerate
- This is dependent on the stellar mass
- We will distinguish between low-mass, moderate-mass, and high-mass evolution

The shell burning law

- This law is shown empirically and numerically, but there are no simple convincing arguments as to why it works (yet)
- It states that "when the region within a burning shell contracts, the region outside the shell expands; and when the region inside the shell expands, the region outside the shell contracts"
- It is also called the mirror principle
- In short: if a core contracts, the envelope expands and vice versa



Credit: JCD, Fig 11.5

First evolution off the main sequence

- As long as there is some radiative region, then one of these equations hold (remember lecture 10):

$$L_{*,\text{low mass MS}} \propto M_*^5 \mu^{7.5}$$

$$L_{*,\text{high mass MS}} \propto M_*^3 \mu^4$$

- Through the Stefan-Boltzmann law, we also have that

$$L_* = 4\pi R_*^2 \sigma T_{\rm eff}^4$$

- A star will revert its evolution from the pre-main sequence stage - as the radius keeps increasing, the effective temperature decreases with luminosity roughly constant or decreasing slightly

Building a convective envelope

- The effective temperature drops till the approach to the Hayashi tracks
- A deep convective envelope has formed
- The radius keeps increasing, but the star can't go into the forbidden zone, hence the luminosity must go up
- This is where the star enters the red giant branch

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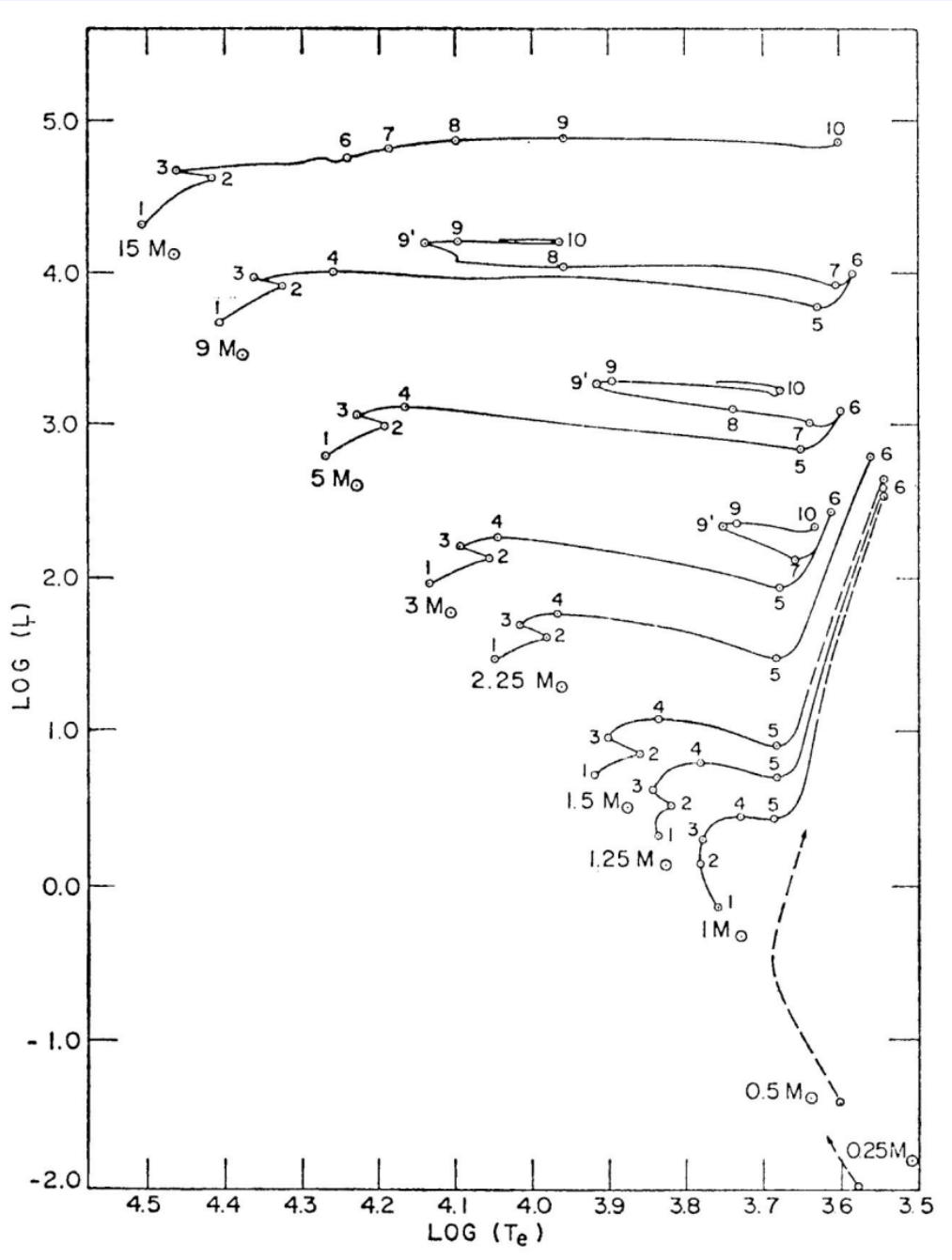

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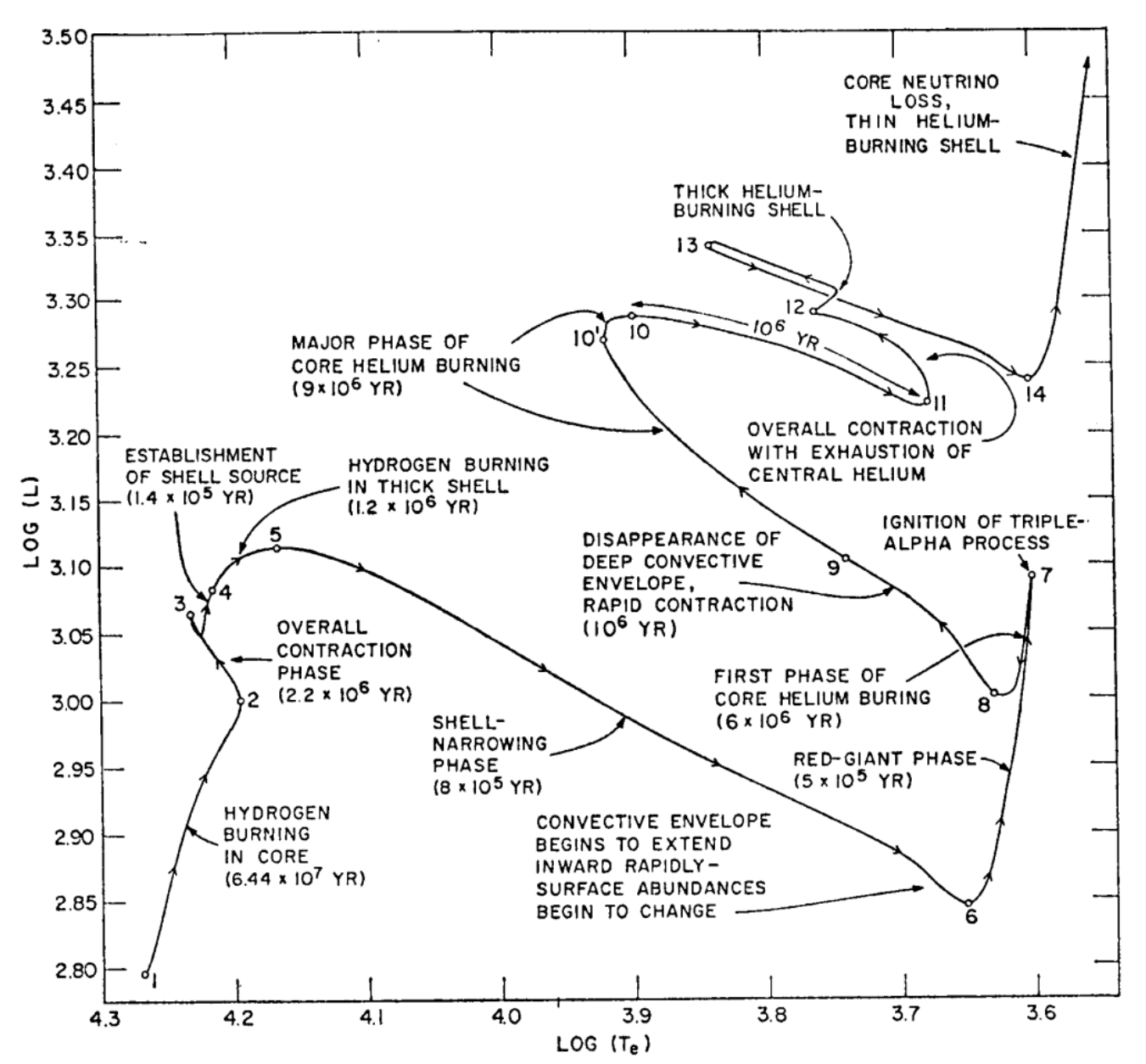
Moderate-mass star evolution - assume 5 $M_\odot$



Credit: JCD, Fig 12.2

Hook in HR diagram

- Upon leaving the main sequence, a "hook" can be seen towards higher effective temperature and luminosity
- This is specific to stars with convective cores (ie all the way down to 1.25 solar mass stars)
- Hydrogen exhausts uniformly (lecture 15) - this decreases energy production
- To compensate, core will heat up by contracting
- This also leads to an increase in effective temperature, creating the "hook"

The Schönberg-Chandrasekhar limit

- We know the non-fusing helium core is isothermal
- There is an upper limit to the mass of an isothermal core that is in hydrostatic and thermal equilibrium
- To be stable, the core mass fraction q_c must be smaller than the Schönberg-Chandrasekhar limit q_{SC}

$$q_c = \frac{m_c}{M} < q_{SC} \approx 0.1$$

- By assuming an ideal gas, we can show that q_{SC} depends on the mean molecular weight ratio of the envelope and the core

The Hertzsprung gap

- Our 5 solar mass star has a helium core mass close to this limit already upon leaving the main sequence
- During H shell fusion, the core grows and quickly exceeds the limit
- The unstable core then contracts on essentially a thermal timescale
- This means the envelope and the full star will expand and the star moves towards the Hayashi track
- This part of the evolution is so fast that we call this region "The Hertzsprung gap" as it is rare to find stars in this region

Summary

- Stars leave the main sequence when H core burning stops
- Inert He cores are isothermal
- H shell burning makes the core contract and the star expand
- When a star reaches the Hayashi track again, it must increase in luminosity: red giant branch
- For a moderate-mass star, the core mass fraction exceeds the Schönberg-Chandrasekhar limit after which the star quickly evolves across the Hertzsprung gap

NEXT LECTURE: He core burning, asymptotic giant branch